Literature review and research hypothesis

The current research mainly examines the economic impact of artificial intelligence (AI) from three aspects: capital and credit markets, corporate behavior, and the labor market. Firstly, AI alters the allocation of credit by improving information processing and risk prediction efficiency, but it may also exacerbate the financing disparities among borrowing groups (Andreas Fuster et al., 2022, Sabrina T Howell et al., 2024). Secondly, corporate AI investment promotes growth through product innovation, while also changing corporate competition and disclosure behaviors (Tania Babina et al., 2024, Elizabeth Blankespoor et al., 2026, Emilio Calvano et al., 2020). Finally, AI can enhance labor productivity, but task automation may also replace some jobs, depress wages, and exacerbate income inequality (Daron Acemoglu and Pascual Restrepo, 2022, David Autor et al., 2024, Erik Brynjolfsson et al., 2025). However, how the risks triggered by AI further affect the availability of household credit remains to be directly tested.

Recent research articles have found that the application of industrial robots , automation and AI has led to a decrease in local wages and employment(Daron Acemoglu et al., 2022, Daron Acemoglu and Pascual Restrepo, 2020). At the same time, the application of AI has reduced the recruitment of non-AI positions in enterprises, and at the same time, data capital will exert a squeezing effect on the share of labor income(Simona Abis and Laura Veldkamp, 2024, Menaka Hampole et al., 2025). Joanne W Hsu et al. (2018)'s research shows that a worker experiencing an unemployment shock or a sudden drop in income can cause a household to default on its home loan. Young borrowers with limited employment histories, savings, and credit records may be particularly sensitive to employment uncertainty. In regions with greater AI exposure, reduced employment opportunities and less stable earnings may weaken applicants' repayment capacity and increase their perceived credit risk. Consequently, lenders may become less willing to approve mortgage applications from young borrowers. Based on this reasoning, this paper proposes the following hypothesis:

H1: In regions with higher exposure to AI, the probability of approval for young people's housing loan applications is lower.

Furthermore, if the labor-displacement effects of AI outweigh productivity-driven employment creation, reductions in labor demand and earnings may negatively affect regional income. Although technological innovation can generate additional income through firm growth, these gains may not fully compensate for the income losses experienced by displaced workers. Therefore, regional income outcomes may depend on the distribution of AI-related gains and losses across workers and firms. This motivates the following hypothesis:

H2: Regions with higher exposure to AI will result in a decrease in the per capita income of that region.

The task substitution theory holds that the development of automation can enable enterprises to complete some tasks originally undertaken by workers at a lower cost, thus reducing the demand for relevant labor (Daron Acemoglu and Pascual Restrepo, 2022). AI can also be regarded as a branch of automation. For occupations with high AI exposure, their work tasks are more likely to be replaced by AI technology, and enterprises may reduce labor costs by reducing recruitment and cutting jobs. While AI may also create new jobs by increasing productivity, task-substitution effects may dominate in the short term, reducing employment in related occupations. Therefore, this paper proposes the following research hypotheses:

H3: The higher the AI exposure level is, the fewer the number of employed people in that region will be.

Table 1 Descriptive Statistics

	N	Mean	SD	P25	Median	P75
approved	1336768	0.812	0.390	1.000	1.000	1.000
ai_exposure	1664252	-0.073	0.261	-0.255	-0.074	0.118
group(derived_race)	1690688	8.217	1.748	8.000	9.000	9.000
loan_type	1690688	1.496	0.809	1.000	1.000	2.000
loan_purpose	1690688	3.567	8.317	1.000	1.000	1.000
income	1648213	74.676	53.556	45.000	62.000	86.000
occupancy_type	1690688	1.043	0.284	1.000	1.000	1.000
per_capita_income_usd	1659784	37366.550	8222.507	31661.000	36376.000	41703.000
Employment	883713	563790.558	708088.801	145700.000	287750.000	697225.000

This article is based on the paper of Jonathan Lee et al. (2026), and controls for county Lender × Year Fixed Effects. This helps to incorporate the changing characteristics of lending institutions over time, such as bank capital, size, risk appetite, management quality, regulatory differences, and business models.

The baseline model is:

$$Approved_{ibct} = \beta AIExposure_i + \gamma X_i + \mu_c + \delta_{bt} + \varepsilon_{ibct}$$

The results are shown in Table 2 . The exposure to county-level AI significantly reduces the likelihood of mortgage applications being approved by young people. Therefore, Hypothesis 1 is valid.

Table2 Baseline results

	(1) approved	(2) approved
ai_exposure	-0.014**	-0.013**
	(-2.42)	(-2.01)
race_cat		0.011***
		(29.99)
loan_type		-0.005***
		(-6.09)
loan_purpose		-0.004***
		(-36.32)
income		0.001***
		(49.91)
occupancy_type		-0.044***
		(-23.05)
construction_method		-0.116***
		(-28.40)
_cons	0.816***	0.872***
	(1993.06)	(143.21)
County_id	Yes	Yes
Bank_id#activity_year	Yes	Yes
Observations	1,290,957	1,290,957
Adjusted R-squared	0.303	0.319

The results in Table 3 indicate that AI exposure will reduce the local per capita income. Therefore, hypothesis H2 is confirmed. And also we can find that AI exposure will reduce the amount of employment(Although the coefficient is not significant, the possible reason is that we were examining the total employment figures rather than those specifically for graduates.)

Table 3 Mechanism

	(1) Per capita income	(2) Employment
ai_exposure	-621.063***	-94526.955
	(-3.50)	(-1.21)
race_cat	-0.770	20.309
	(-1.34)	(1.13)
loan_type	22.269***	616.985***
	(8.66)	(2.69)
loan_purpose	-2.075***	-25.907
	(-7.25)	(-0.78)
income	0.276***	2.329
	(9.16)	(1.42)
occupancy_type	-8.275**	-221.455**
	(-2.45)	(-2.49)
construction_method	-10.987**	145.422
	(-2.23)	(0.73)
_cons	37301.087***	567446.539***
	(2255.40)	(122.98)
County_id	Yes	Yes
Bank_id#activity_year	Yes	Yes
Observations	1,615,644	856382
Adjusted R-squared	0.992	0.999

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